COMP1531
Software Engineering Fundamentals

Week 9

Projects - Deployment

In This Lecture

Why?

- The purpose of most software is for **people to use it**, and for that to happen we need processes to make it available for usage

What?

- Deployment
- Continuous Delivery
- Continuous Deployment
- DevOps

Deployment

Activities relating to making a software system available for use

Many diagrams in this lecture are sourced from both gitlab and atlassian

History Of Deployment

- Historically, deployment was much **less frequent** and **much more physical**.
- Code would be worked on for days at a time without being tested, and **deployed sometimes years apart**. This was largely due to software historically being a physical asset (e.g. a CD).

Something Changed

However, that isn't the case anymore. Two major changes have occurred over the last 10-15 years:

- Increased prevalence of **web-based apps** (no local installs)
- Improvement to **internet connectivity, speed, bandwidth**

These changes (and more) have allowed for the pushing of updated software to users to be substantially more frequent. E.G. Think about how quickly Netflix is updated.

A movement from software as an asset, to software as a service, has catalysed this transition

Cloud Services

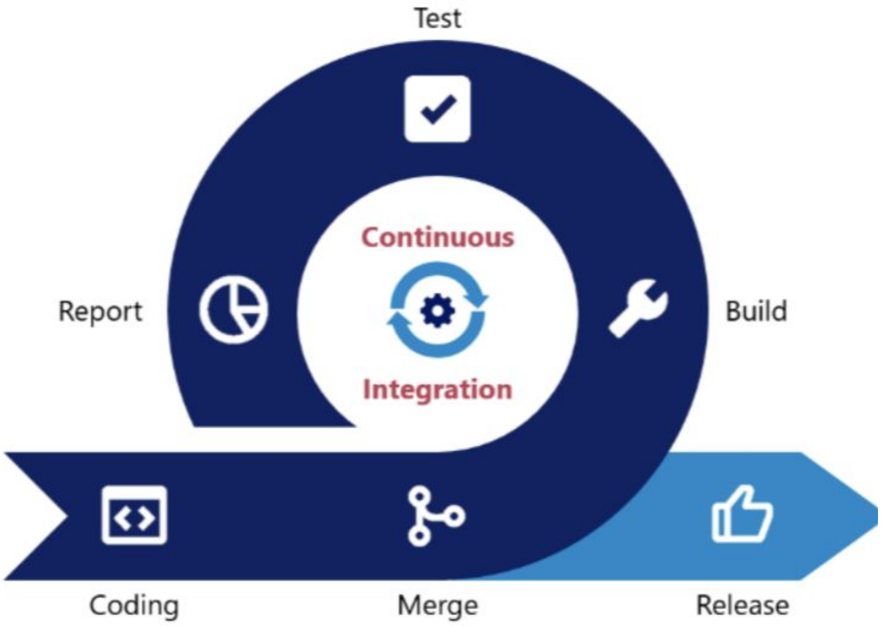
Numerous cloud services offer the ability to "easily" deploy your web applications.

Examples include:

- Amazon Web Services
- Google App Engine
- Vercel

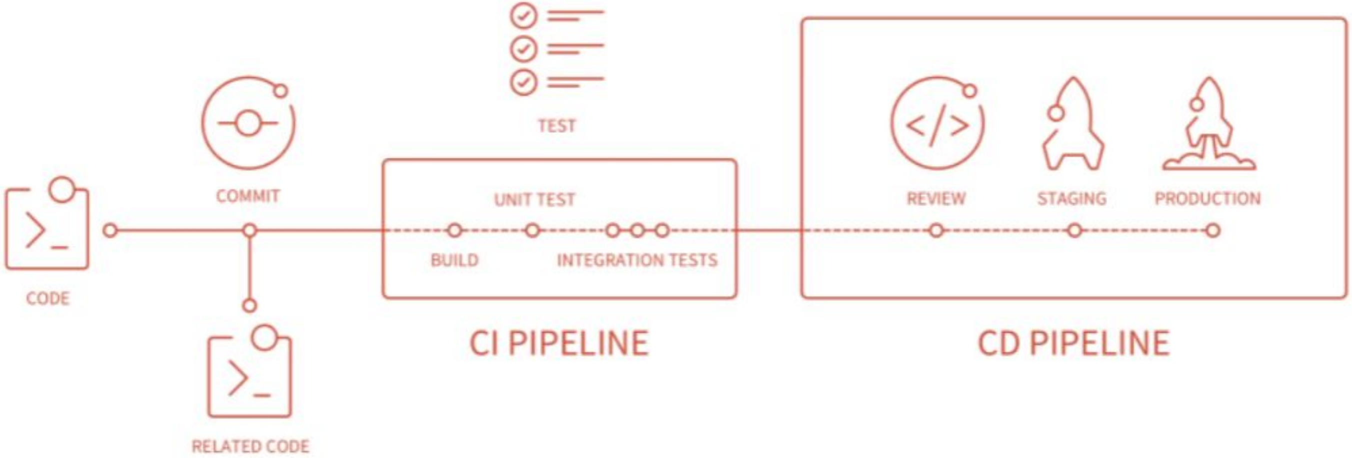
Recap: Continuous Integration

Continuous integration: Practice of automating the integration of code changes from multiple contributors into a single software project.



Modern Deployment With CI

To achieve rapid deployment cycles, modern deployment isn't as simple as pushing code. Rather, a heavily integrated and automated approach is preferred. This new part of the pipeline we will explore is called **Continunous Delivery.**



Source: Gitlab

Continuous Delivery

Continuous delivery: Allows accepted code changes to be deployed to customers quickly and sustainably.

This involves the automation of the release process such that releases can be done in a "button push".

Continuous Delivery

Many companies will have a daily or weekly deployment or "ship".

To "ship" code t**here is a "sign off" process** (human intervention) to deploy the code.

Release Methods

Continuous delivery is often concerned with more than just going from "your computer" to a "production environment".

We can have 0 or more interim stages of release.

Release Methods

For example, you might have 3 core tiers:

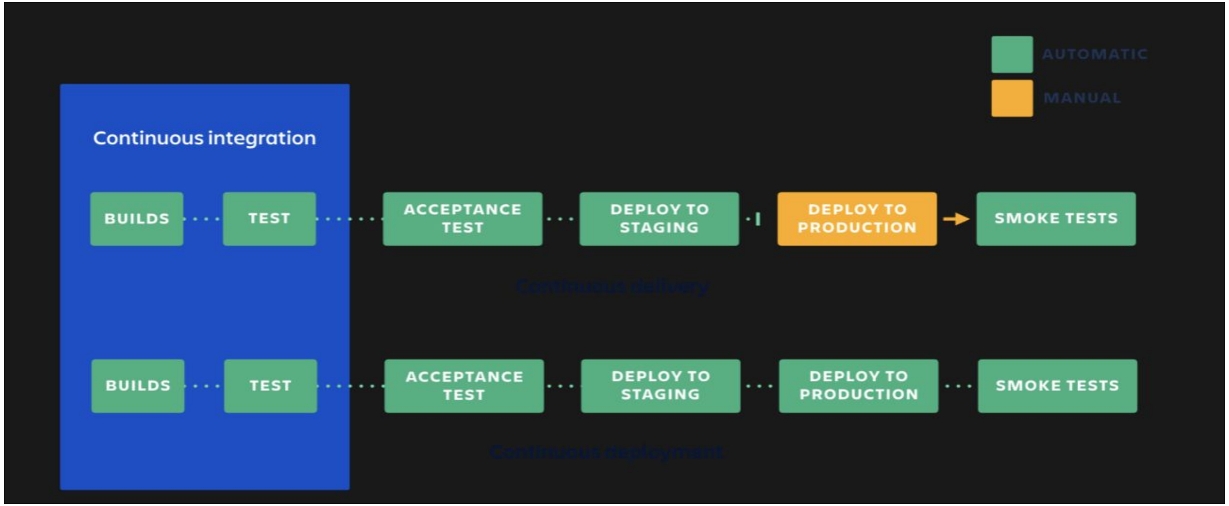
- dev: Released often, available to developers to see their changes in deployment
- test/staging: As close to release as possible, ideally identical to prod
- prod: Released to customers, ideally as quickly as possible

As you move down the stages, things tend to be more stable.

Continuous Deployment

Continuous Deployment is an **extension of Continuous Delivery** whereby changes attempt to flight toward production automatically, and the only thing stopping them is a failed test.

Source: Atlassian



Deploying On Your Own

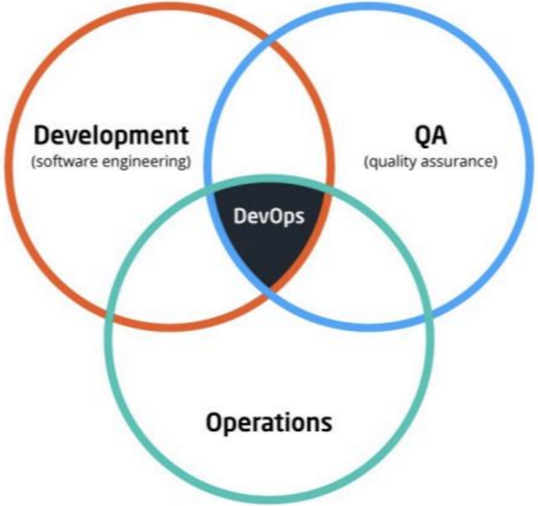
This term COMP1531 use a free service known as "Vercel" to let students deploy their backend to the cloud.

Instructions of how to set this up are found in lab10_deploy.

DevOps

A decade ago, you would typically expect people from a multitude of separate roles to work together to coordinate the deployment of work.

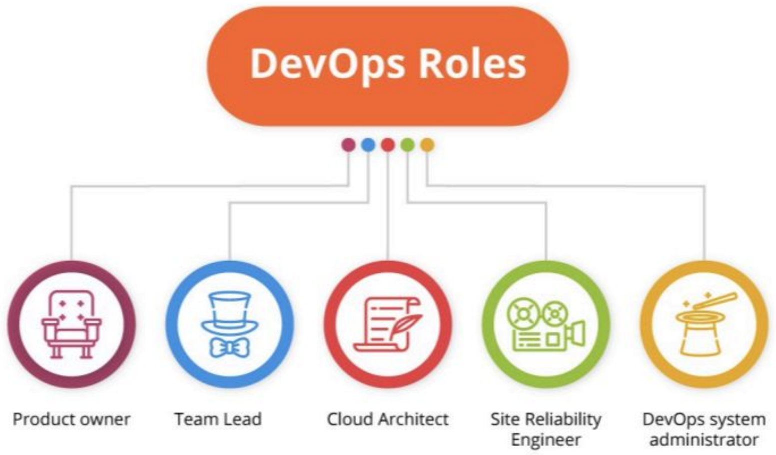
In recent years, DevOps as a role has become a more relevant role that focuses on skill(s) that include an overlap of development, deployment, and quality assurance.



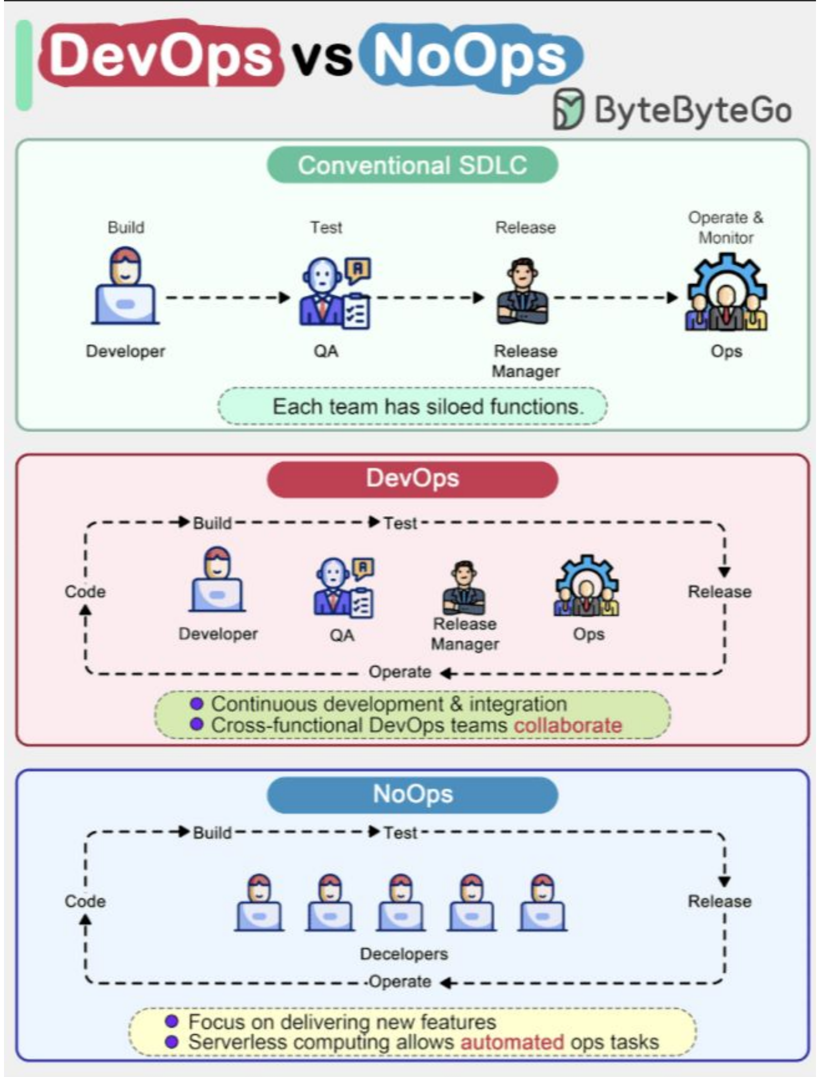
DevOps

Wikipedia says that DevOps is a set of practices intended to **reduce the time between committing a change to a system and the change being placed into normal production**, while ensuring high quality.

As development teams become less silo'ed, modern DevOps is less a role, and more a series of roles or aspect of a role.



[Source & Reading]


DevOps vs NoOps
ByteByteGo
Conventional SDLC
Build
Test
Release
Operate & Monitor
Developer
QA
Release Manager
Ops
Each team has siloed functions.
DevOps
Build
Test
Code
Release
Developer
QA
Release Manager
Ops
Operate
Continuous development & integration
Cross-functional DevOps teams collaborate
NoOps
Build
Test
Code
Release
Decelopers
Operate
Focus on delivering new features
Serverless computing allows automated ops tasks

Maintenance & Monitoring

Maintenance: After deployment, the use of analytics and monitoring tools to ensure that as the platform is used and remains in a healthy state.

Monitoring often has two purposes:

1. **Preserving user experience**: Monitoring errors, warnings, and other issues that affect performance or uptime.
2. **Enhancing user experience**: Using analytical tools to monitor users or understanding their interactions. Often leads to customer interviews and user stories.

DevOps / Site Reliability Engineers

Health is defined by developers, but often consists of:

- Monitoring 4XX and 5XX errors (user did something wrong e.g. 404 (page not found), 400 (bad request) OR system is broken e.g. 500 (server crash), 503 (service unavailable)
- Ensuring disk, memory, cpu, and network is not overloaded

Often these aren't actively monitored, but rather monitored with alerts and triggers

Further Reading

- Atlassian Continuous Delivery
- Gitlab Continuous Integration
- Atlassian Delivery vs Deployment

Feedback

Or go to the form here